

Assignment 4

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R Markdown

```
library(ISLR2)
```

```
## Warning: package 'ISLR2' was built under R version 4.2.3
```

13. This question should be answered using the Weekly data set, which is part of the ISLR2 package. This data is similar in nature to the Smarket data from this chapter's lab, except that it contains 1,089 weekly returns for 21 years, from the beginning of 1990 to the end of 2010.

- (a) Produce some numerical and graphical summaries of the Weekly data. Do there appear to be any patterns?

```
data <- data("Weekly")
str(Weekly)
```

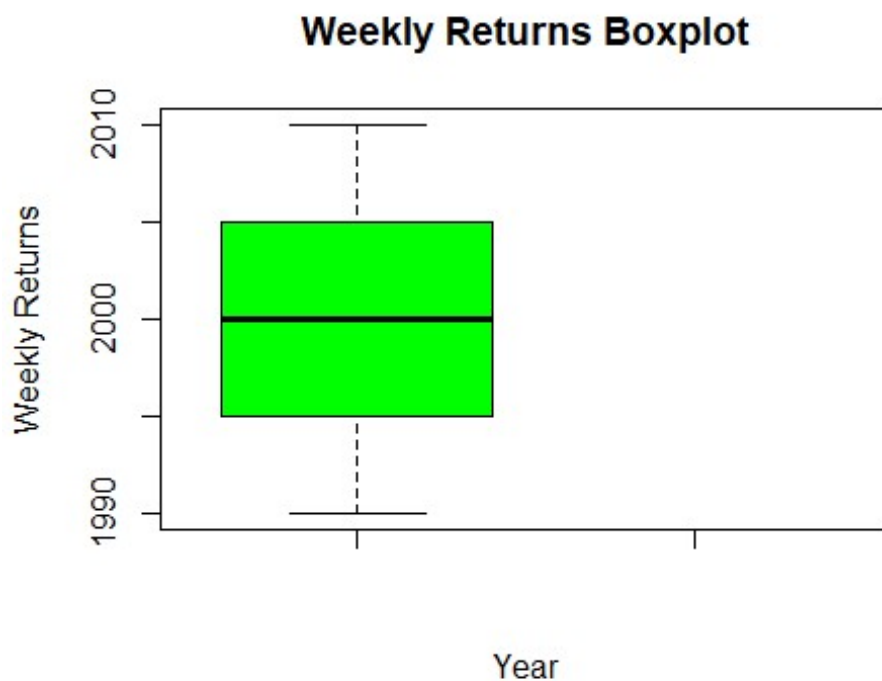
```
## 'data.frame': 1089 obs. of 9 variables:
## $ Year : num 1990 1990 1990 1990 1990 1990 1990 1990 1990 1990 1990 ...
## $ Lag1 : num 0.816 -0.27 -2.576 3.514 0.712 ...
## $ Lag2 : num 1.572 0.816 -0.27 -2.576 3.514 ...
## $ Lag3 : num -3.936 1.572 0.816 -0.27 -2.576 ...
## $ Lag4 : num -0.229 -3.936 1.572 0.816 -0.27 ...
## $ Lag5 : num -3.484 -0.229 -3.936 1.572 0.816 ...
## $ Volume : num 0.155 0.149 0.16 0.162 0.154 ...
## $ Today : num -0.27 -2.576 3.514 0.712 1.178 ...
## $ Direction: Factor w/ 2 levels "Down","Up": 1 1 2 2 2 1 2 2 2 1 ...
```

```
summary(Weekly)
```

```
##      Year      Lag1      Lag2      Lag3
## Min.   :1990   Min.   :-18.1950   Min.   :-18.1950   Min.   :-18.1950
## 1st Qu.:1995   1st Qu.: -1.1540   1st Qu.: -1.1540   1st Qu.: -1.1580
## Median :2000   Median :  0.2410   Median :  0.2410   Median :  0.2410
## Mean   :2000   Mean   :  0.1506   Mean   :  0.1511   Mean   :  0.1472
## 3rd Qu.:2005   3rd Qu.:  1.4050   3rd Qu.:  1.4090   3rd Qu.:  1.4090
## Max.   :2010   Max.   : 12.0260   Max.   : 12.0260   Max.   : 12.0260
##      Lag4      Lag5      Volume      Today
## Min.   :-18.1950   Min.   :-18.1950   Min.   :0.08747   Min.   :-18.1950
## 1st Qu.: -1.1580   1st Qu.: -1.1660   1st Qu.:0.33202   1st Qu.: -1.1540
## Median :  0.2380   Median :  0.2340   Median :1.00268   Median :  0.2410
## Mean   :  0.1458   Mean   :  0.1399   Mean   :1.57462   Mean   :  0.1499
## 3rd Qu.:  1.4090   3rd Qu.:  1.4050   3rd Qu.:2.05373   3rd Qu.:  1.4050
## Max.   : 12.0260   Max.   : 12.0260   Max.   :9.32821   Max.   : 12.0260
## Direction
```

```
## Down:484
## Up :605
##
##
##
##
```

```
boxplot(Weekly$Year, Weekly$Today_return, xlab = "Year", ylab = "Weekly
Returns", col = "green", main = "Weekly Returns Boxplot")
```



- (b) Use the full data set to perform a logistic regression with Direction as the response and the five lag variables plus Volume as predictors. Use the summary function to print the results. Do any of the predictors appear to be statistically significant? If so, which ones?

```
glm.fits <- glm(
  Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume, data = Weekly,
  family = binomial)
```

```
summary(glm.fits)
```

```
##
## Call:
## glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
##      Volume, family = binomial, data = Weekly)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -1.6949 -1.2565 0.9913 1.0849 1.4579
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.26686    0.08593   3.106  0.0019 **
## Lag1        -0.04127    0.02641  -1.563  0.1181
## Lag2         0.05844    0.02686   2.175  0.0296 *
## Lag3        -0.01606    0.02666  -0.602  0.5469
## Lag4        -0.02779    0.02646  -1.050  0.2937
## Lag5        -0.01447    0.02638  -0.549  0.5833
## Volume      -0.02274    0.03690  -0.616  0.5377
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1496.2  on 1088  degrees of freedom
## Residual deviance: 1486.4  on 1082  degrees of freedom
## AIC: 1500.4
##
## Number of Fisher Scoring iterations: 4
```

In our summary output, we can see that for variables Lag 2 is significantly smaller than our p value of 0.05 which indicate it is statistically significant predictor of the reposne variable “direction” $\Pr(>|z|) = 0.0296$.

- (c) Compute the confusion matrix and overall fraction of correct predictions. Explain what the confusion matrix is telling you about the types of mistakes made by logistic regression.

```
names(Weekly)

## [1] "Year"      "Lag1"      "Lag2"      "Lag3"      "Lag4"      "Lag5"
## [7] "Volume"    "Today"     "Direction"

glm.probs <- predict(glm.fits, type = "response")
glm.probs[1:10]

##           1           2           3           4           5           6           7
8
## 0.6086249 0.6010314 0.5875699 0.4816416 0.6169013 0.5684190 0.5786097
0.5151972
##           9          10
## 0.5715200 0.5554287

glm.pred <- rep("Down", 1250)
glm.pred[glm.probs > 0.5] = "Up"

#table(glm.pred, Weekly$Direction)
```

The confusion matrix for logistic regression is used to assess the performance of a model in predicting binary outcome and it is broken down into 4 categories

-True positive (correctly predicted positive outcome) -True negatives(correctly predicted negative outcomes) -False positive(incorrectly predicted positive outcomes) -False negative(incorrectly predicted false outcomes)

The model makes mistakes and predictions are only correct sometimes and must be noted.

- (d) Now fit the logistic regression model using a training data period from 1990 to 2008, with Lag2 as the only predictor. Compute the confusion matrix and the overall fraction of correct predictions for the held out data (that is, the data from 2009 and 2010).

```
train_data <- subset(Weekly, Year >= 1990 & Year <= 2008)
test_data <- subset(Weekly, Year >= 2009)

model <- glm(Direction ~ Lag2, data = train_data, family = binomial)
predictions <- predict(model, newdata = test_data, type = "response")

predicted_classes <- ifelse(predictions > 0.5, "Up", "Down")
conf_matrix <- table(predicted_classes, test_data$Direction)

conf_matrix

##
## predicted_classes Down Up
##           Down    9   5
##           Up    34  56

accuracy <- sum(diag(conf_matrix)) / sum(conf_matrix)
accuracy

## [1] 0.625

library(MASS)

##
## Attaching package: 'MASS'

## The following object is masked from 'package:ISLR2':
##
##      Boston

lda.fit <- lda(Direction ~ Lag2, data = train_data)
lda.pred <- predict(lda.fit, newdata = test_data)

lda_conf_matrix <- table(lda.pred$class, test_data$Direction)
lda_conf_matrix
```

```
##
##           Down Up
##   Down      9  5
##   Up       34 56

lda_accuracy <- sum(diag(lda_conf_matrix)) / sum(lda_conf_matrix)
lda_accuracy

## [1] 0.625

qda.fit <- qda(Direction ~ Lag2, data = train_data)
qda.pred <- predict(qda.fit, newdata = test_data)
qda_conf_matrix <- table(qda.pred$class, test_data$Direction)
qda_conf_matrix

##
##           Down Up
##   Down      0  0
##   Up       43 61

qda_accuracy <- sum(diag(qda_conf_matrix)) / sum(qda_conf_matrix)
qda_accuracy

## [1] 0.5865385

library(class)

knn_model <- knn(train = train_data[, "Lag2", drop = FALSE],
                 test = test_data[, "Lag2", drop = FALSE],
                 cl = train_data$Direction,
                 k = 1)

knn_conf_matrix <- table(knn_model, test_data$Direction)
knn_conf_matrix

##
## knn_model Down Up
##       Down  21 30
##       Up   22 31

knn_accuracy <- sum(diag(knn_conf_matrix)) / sum(knn_conf_matrix)
knn_accuracy

## [1] 0.5

library(e1071)

## Warning: package 'e1071' was built under R version 4.2.3

nb_model <- naiveBayes(Direction ~ Lag2, data = train_data)
nb_predictions <- predict(nb_model, newdata = test_data)
```

```

nb_conf_matrix <- table(nb_predictions, test_data$Direction)

nb_conf_matrix

##
## nb_predictions Down Up
##           Down    0  0
##           Up    43 61

nb_accuracy <- sum(diag(nb_conf_matrix)) / sum(nb_conf_matrix)
nb_accuracy

## [1] 0.5865385

```

16. Using the Boston data set, fit classification models in order to predict whether a given census tract has a crime rate above or below the median. Explore logistic regression, LDA, naive Bayes, and KNN models using various subsets of the predictors. Describe your findings. Hint: You will have to create the response variable yourself, using the variables that are contained in the Boston data set.

```

data(Boston)
library(MASS)
library(e1071)
library(class)
library(caTools)

## Warning: package 'caTools' was built under R version 4.2.3

str(Boston)

## 'data.frame':    506 obs. of  14 variables:
## $ crim      : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
## $ zn        : num  18 0 0 0 0 12.5 12.5 12.5 12.5 ...
## $ indus     : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
## $ chas      : int   0 0 0 0 0 0 0 0 0 0 ...
## $ nox       : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524
0.524 ...
## $ rm        : num  6.58 6.42 7.18 7 7.15 ...
## $ age       : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
## $ dis       : num  4.09 4.97 4.97 6.06 6.06 ...
## $ rad       : int   1 2 2 3 3 3 5 5 5 5 ...
## $ tax       : num  296 242 242 222 222 222 311 311 311 311 ...
## $ ptratio   : num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
## $ black     : num  397 397 393 395 397 ...
## $ lstat     : num  4.98 9.14 4.03 2.94 5.33 ...
## $ medv      : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...

Boston$CrimeAboveMedian <- ifelse(Boston$crim > median(Boston$crim), "Above",
"Below")

set.seed(123)
split <- sample.split(Boston$CrimeAboveMedian, SplitRatio = 0.7)

```

```

train_data <- subset(Boston, split == TRUE)
test_data <- subset(Boston, split == FALSE)

table(train_data$CrimeAboveMedian)

##
## Above Below
## 177 177

train_data$CrimeAboveMedian <- ifelse(train_data$CrimeAboveMedian == "Above",
1, 0)

```

Logistic Regression

```

log_model <- glm(CrimeAboveMedian ~ ., data = train_data, family = binomial)

## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

log_pred <- predict(log_model, newdata = test_data, type = "response")
log_pred_class <- ifelse(log_pred > 0.5, "Above", "Below")

```

LDA

```

# LDA
lda_model <- lda(CrimeAboveMedian ~ ., data = train_data)
lda_pred <- predict(lda_model, newdata = test_data)$class

```

Naive Bayes

```

nb_model <- naiveBayes(CrimeAboveMedian ~ ., data = train_data)
#nb_pred <- predict(nb_model, newdata = test_data)$class

```

KNN model

```

#knn_model <- knn(train = train_data[, -c(1, 14)], test = test_data[, -c(1,
14)], cl = #train_data$CrimeAboveMedian, k = 5)

log_accuracy <- mean(log_pred_class == test_data$CrimeAboveMedian)
lda_accuracy <- mean(lda_pred == test_data$CrimeAboveMedian)
#nb_accuracy <- mean(nb_pred == test_data$CrimeAboveMedian)
knn_accuracy <- mean(knn_model == test_data$CrimeAboveMedian)

## Warning in `==.default`(knn_model, test_data$CrimeAboveMedian): longer
object
## length is not a multiple of shorter object length

## Warning in is.na(e1) | is.na(e2): longer object length is not a multiple
of
## shorter object length

log_accuracy

```

```
## [1] 0.9802632
```

```
lda_accuracy
```

```
## [1] 0
```

```
nb_accuracy
```

```
## [1] 0.5865385
```

```
knn_accuracy
```

```
## [1] 0
```